

Changing Mortality rate in Canada: Better Healthcare or Demographic Change?

Disclaimer: I have taken the help of AI to fix the code for my analysis in Python. I have also used it to find grammatical mistakes in my writing. But all writing and analysis was completely mine. Also I used it to find possible issues in my argument in the research as a 3rd person perspective so that I can Improve the gaps in my research.

Introduction

All the countries around the world treat reduction in mortality rate as a measurement of improvement in the Healthcare sector. In Canada, the mortality rate is also attributed to improvement in the health sector. From 2000-2019 the mortality rate in Canada fluctuated between 7.0-7.6 per 1000 population but from then it increased significantly from 2020-2024, at one point up to 8.7 per 1000 which can be attributed to COVID pandemic. We saw a very little change in mortality rate over the years. Because mortality rate can be influenced by actual healthcare improvement or population dynamics change such as age, education, lifestyle or earnings. In this paper, we will test which of the indicators can be attributed to the actual improvement of healthcare.

‘Avoidable Mortality’ is defined as “untimely deaths that should not occur in the presence of timely and effective health care, including prevention”. ‘Avoidable Mortality’ consists of two subgroups: ‘Preventable mortality’ and ‘Treatable mortality’. “Preventable mortality” refers to deaths that could be prevented through primary preventative actions in incidence reduction, such as vaccine, smoking reduction and seatbelt use. ‘Treatable mortality’ refers to deaths that could

be treated through secondary and tertiary prevention efforts, such as effective screening and treatment of an existing disease. Treatable Mortality is used as a proxy for healthcare performance because it can properly indicate if mortality rate decreased only if the quality and size of the healthcare sector has improved significantly. In this paper we only consider the mortality that could have been prevented via proper medical treatment defined as ‘treatable mortality’ because it can be properly accounted for, unlike ‘Preventable Mortality’. We ignore deaths that cannot be prevented through proper healthcare such as old age or conditions without an available cure.

Literature Review

The mortality rate in Canada over the last few decades saw a significant decline in ‘Treatable Mortality’. In Canada, avoidable mortality decreased by 46.9% between 1975 and 1999 (James, 2006). Though absolute mortality related to disease has declined but impact of the healthcare system is still questionable as the overall mortality rate changes very little. The concept of Avoidable Mortality originated in 1976 when Rutstein suggested measuring quality of care through untimely deaths which should not occur in the presence of timely and good quality care (Kossarova, 2013). James (2006) showed in his paper, In Canada, Avoidable Mortality has decreased considerably and regional differences have narrowed, although there continues to exist important regional disparities. But his study lacks recent data as it was conducted from 1976 to 1999. Also we see that from 2000-2004, the mortality rate has increased somewhat instead of decreasing. Recent studies (1991–2016) shows that although premature mortality rates declined in most groups, the relative risk of premature mortality for the lowest income quintile increased significantly from 2.10 to 2.79 among men and 1.72 to 2.50 among women. This shows that the

benefits of healthcare advancements are not being shared equally, with socioeconomically disadvantaged groups falling behind (Shahidi, 2020). Mackenbach (2013) tried a different approach in his paper that identified 14 conditions with declining mortality linked to healthcare interventions but validated findings through a Delphi panel of experts which is taking the opinions of experts in the healthcare field rather than empirical data. My study fills the gap by empirically testing whether mortality differences across Canadian provinces reflect healthcare performance or population characteristics.

Data

This paper uses panel data for the ten Canadian provinces over the period 2001–2023 with 230 province-year observations. The dependent variable is ‘treatable mortality’ data which is age-standardized using the 2021 population. The paper uses ICD-10 for classification of diseases. ICD-10 stands for International Classification of Diseases, 10th revision. It is the global standard system used by every country to classify and code causes of death consistently. Canada adopted ICD-10 for mortality reporting death statistics in 2000. We ignored data before 2000 as the definition of what should be considered as treatable mortality changed from ICD-9 to ICD-10 in Canada. According to ICD-10, “Treatable (or amenable) mortality refers to deaths that, in the light of medical knowledge and technology at the time of death, could be mainly avoided through timely and effective health care interventions, including secondary prevention and treatment among individuals aged less than 75 years.” but we only use the data till age 65 or less so that we have consistent available data for all variables.

There are 55 causes of death in the ICD-10 amenable mortality list. Including all 55 would mix healthcare system effects with factors the system cannot control, such as road accidents, homicides, Alzheimer's disease, and lung cancer driven by patients' smoking rates. Ten causes of death are therefore selected that cover the main dimensions of health system performance:

Table 1: ICD-10 conditions included in the treatable mortality measure.

Condition	What a death signals
Diabetes (E10–E14)*	Poor chronic disease management and monitoring
Hypertensive disease (I10–I13)	Undiagnosed or unmanaged blood pressure
Ischaemic heart disease (I20–I25)	Access to cardiac surgery and intervention
Cerebrovascular disease (I60–I69)	Speed of stroke response and treatment
Influenza and pneumonia (J09–J18)	Gaps in vaccination and primary care
Appendicitis (K35–K38)	Access to emergency surgical care
Breast cancer (C50)	Screening program coverage and quality
Cervical cancer (C53)	Pap screening program effectiveness
Colorectal cancer (C18–C21)	Early detection through screening
Perinatal causes (P00–P96)	Quality of obstetric and neonatal care

**codes represents the ICD-10 disease code.*

I have used 5 independent variables which capture population dynamics along with healthcare supply. I collected Physicians per 100,000 population from the Canadian Institute for Health Information (CIHI). I use it as a primary proxy for healthcare supply. Per capita provincial government health expenditure, measured in 2010 constant dollars, is collected from the CIHI National Health Expenditure Database (NHEX). This variable captures healthcare spending

intensity which physician supply alone cannot. Median household income (in 2023 constant dollars), Population aged 65 and over is converted to a percentage, share of the population with postsecondary education is collected from Statistics Canada. These variables capture the demographic data of each province.

Regression Methodology

Model Equation:

$$\text{treatable mortality}_{it} = \beta_1 \cdot \text{physicians per 100k}_{it} + \beta_2 \cdot \text{median income}_{it} + \beta_3 \cdot \text{percentage 65 plus}_{it} + \beta_4 \cdot \text{spending per capita}_{it} + \beta_5 \cdot \text{education}_{it} + \mu_{it} + \gamma_{it} + \varepsilon_{it}$$

treatable mortality_{it} : Deaths per 100,000 population, province i, year t.

physicians per100k_{it} : Physicians per 100,000 population, province i, year t.

median income_{it} : Median household income, province i, year t.

percentage 65 plus_{it} : % Population aged 65 or more, province i, year t.

spending per capita_{it} : Provincial government health expenditure per capita, province i, year t.

education_{it} : % Population aged 15+ with a postsecondary certificate, diploma, or university degree, province i, year t.

μ_{it} : Province Fixed Effects.

γ_{it} : Year Fixed Effects.

ε_{it} : Error term.

This specification uses a two-way fixed effects panel OLS model with cluster-robust standard errors clustered at the province level. My two explanatory variables of primary interest are physicians per 100,000 and per capita health spending, both of which proxy for the size and quality of the healthcare system. Median income and the share of the population with

postsecondary education serve as socioeconomic controls and the share of the population aged 65 or more captures demographic dynamics. Province fixed effects absorb time-invariant differences across provinces (geography, culture etc.), while year fixed effects absorb Canada-wide shocks common to all provinces in a given year (national policy changes, economic cycles, the COVID-19 pandemic). Standard errors are clustered at the province level to account for serial correlation within each province over time.

Results

We can see in Figure 1 below that the average treatable mortality rate varies across the ten provinces over the 2001–2023 period in Canada. Newfoundland and Labrador has the highest average rate at approximately 550 deaths per 100,000, Alberta records the lowest rate at roughly 340 per 100,000 people which is about 60% lower than Newfoundland. The Atlantic provinces consistently have the highest mortality rates in Canada, while Alberta and Quebec record the lowest. But, these averages can be misleading because provinces have different demographics. By 2023, approximately 19.6% of Newfoundland's population was aged 65 or over, compared to only 13.1% in Alberta.

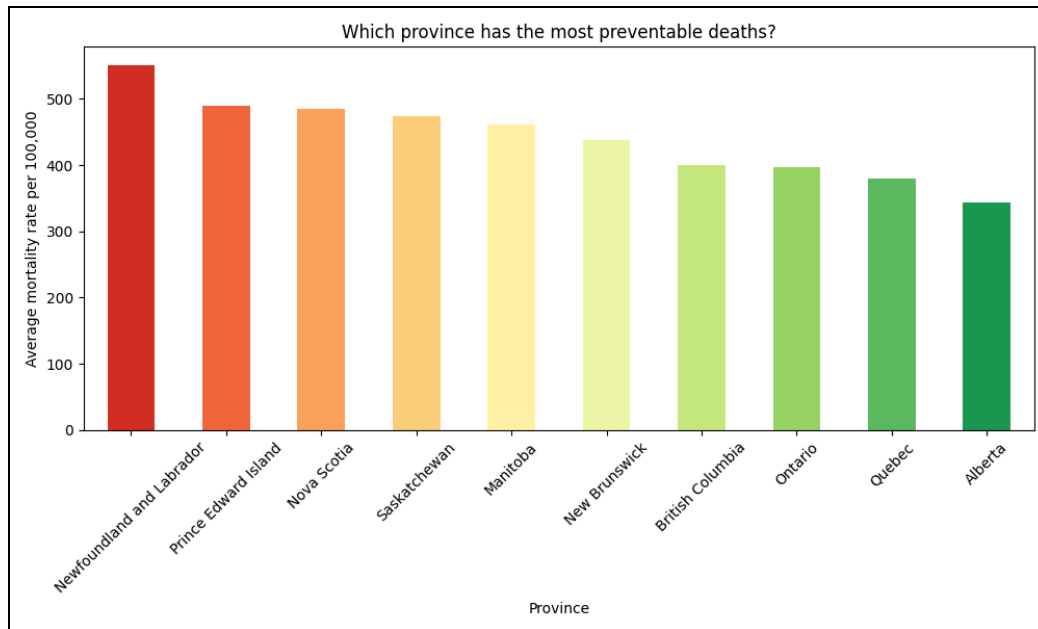


Figure 1: Cross-Provincial Patterns

Figure 2 shows that every province improved over the period 2001–2023, with mortality rates falling in all provinces. But the gap between provinces has not decreased. Newfoundland started at around 590 deaths per 100,000 in 2001 and remains above 500 in 2023, while Alberta has dropped below 300. Prince Edward Island shows the largest year-to-year fluctuations, possibly due to its small population size.

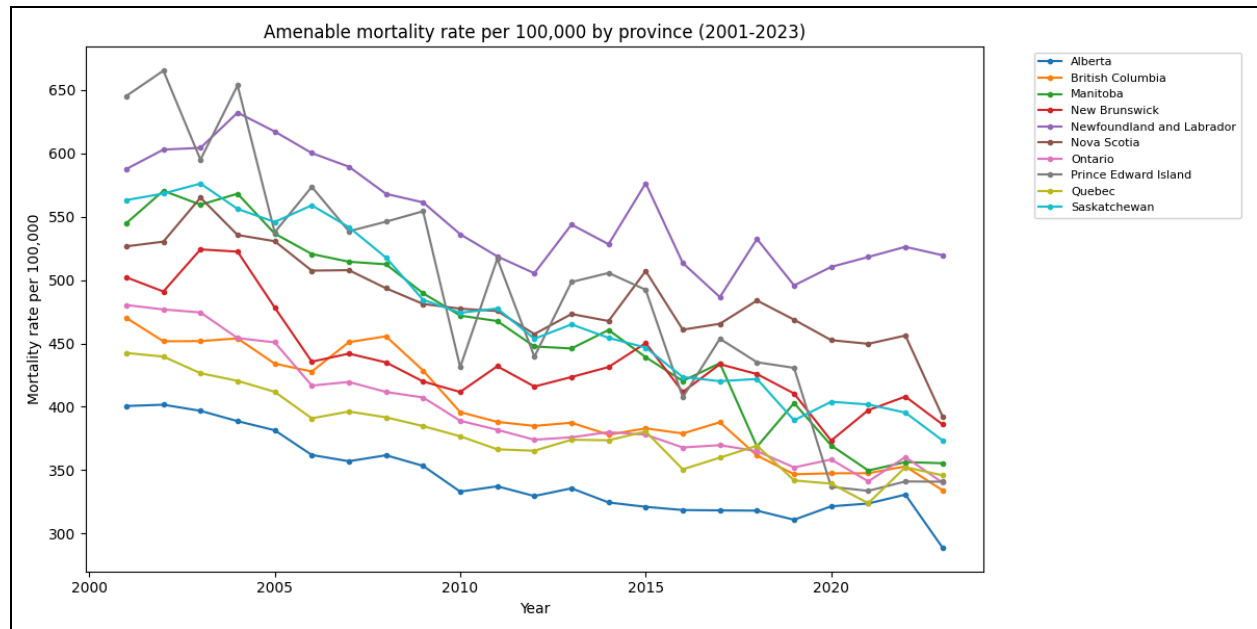


Figure 2: Mortality Trends over Time

Figure 3 plots mortality against real median income. We can see a clear pattern as provinces with higher income tend to have lower treatable mortality. But this simple relationship does not prove that income alone causes lower mortality. Higher-income provinces also tend to have younger populations, more doctors, and different demographics. To separate the true effect of income from these other factors, we run the regression.

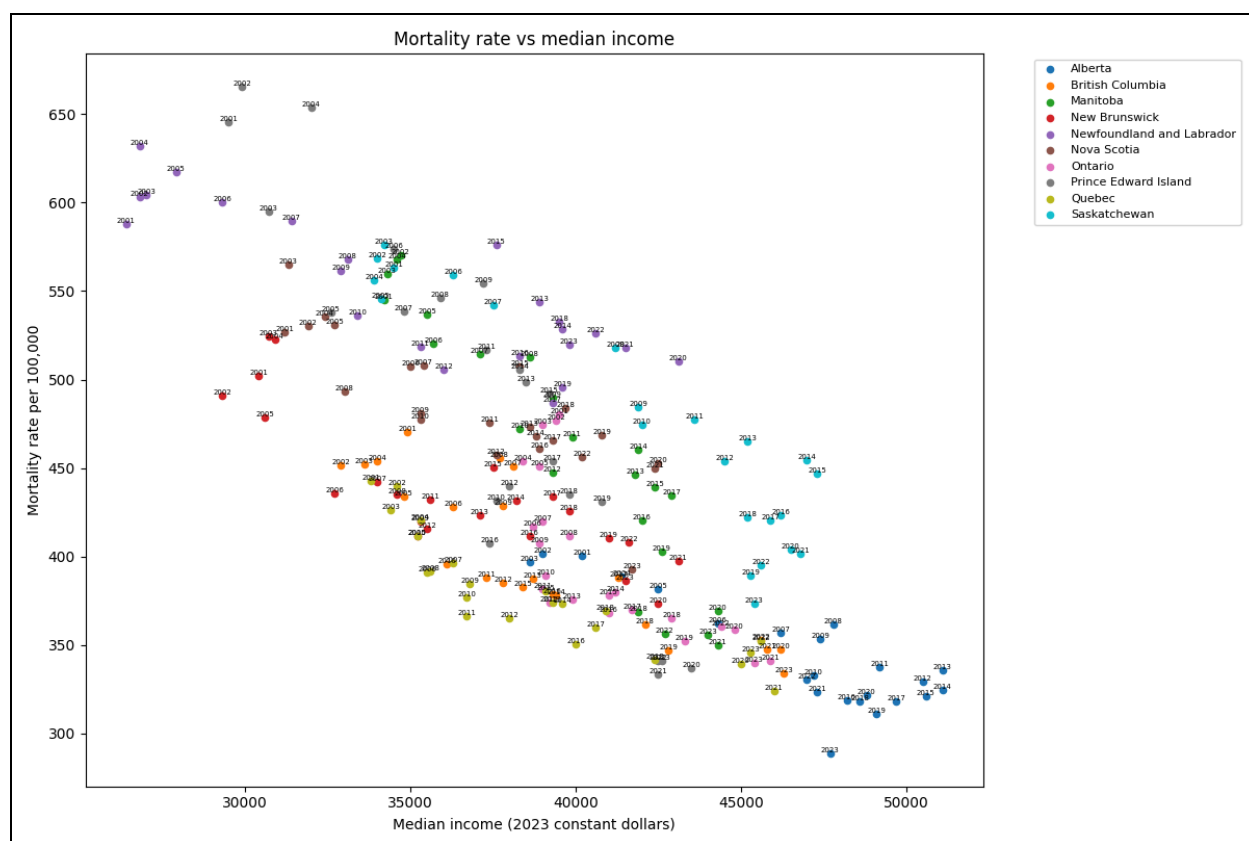


Figure 3: Mortality vs Real Median Income

Regression Results:

Table 2 presents the two-way fixed effects estimates using cluster-robust standard errors at the province level. Firstly, within the five independent variables, only the share of population aged 65 and over is economically meaningful, though only marginally significant at the 10% level. A one percentage-point increase in the 65+ share is associated with approximately 10.93 ($p = 0.061$) additional treatable deaths per 100,000. So demographic change is the main determinant in mortality change. Physicians per 100,000 have a coefficient of 0.30 ($p = 0.48$), and per capita government health spending carries a coefficient of 0.02 ($p = 0.34$) meaning both measures of

healthcare capacity are statistically insignificant. The two socioeconomic controls are also found to be insignificant. Median household income has the expected negative sign ($\beta = -0.0034$, $p = 0.15$), and the share of the population with postsecondary education also carries a negative coefficient ($\beta = -1.91$, $p = 0.40$). This happens because both income and education change slowly within provinces over time so the fixed effects absorb slow-moving within-province change.

PanelOLS Estimation Summary						
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Dep. Variable:	mortality_rate	R-squared:		0.1691		
Estimator:	PanelOLS	R-squared (Between):		0.2624		
No. Observations:	230	R-squared (Within):		-0.3559		
Date:	Mon, Apr 20 2026	R-squared (Overall):		0.2535		
Time:	18:51:12	Log-likelihood		-1051.3		
Cov. Estimator:	Clustered					
		F-statistic:		7.8541		
Entities:	10	P-value		0.0000		
Avg Obs:	23.000	Distribution:		F(5,193)		
Min Obs:	23.000					
Max Obs:	23.000	F-statistic (robust):		9.6536		
		P-value		0.0000		
Time periods:	23	Distribution:		F(5,193)		
Avg Obs:	10.0000					
Min Obs:	10.0000					
Max Obs:	10.0000					
Parameter Estimates						
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	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI

physicians_per100k	0.3000	0.4270	0.7026	0.4831	-0.5422	1.1423
median_income	-0.0034	0.0023	-1.4630	0.1451	-0.0080	0.0012
pct_65plus	10.932	5.8108	1.8813	0.0614	-0.5287	22.393
spending_percapita	0.0208	0.0217	0.9601	0.3382	-0.0219	0.0635
education	-1.9079	2.2635	-0.8429	0.4003	-6.3722	2.5565
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F-test for Poolability: 10.188						
P-value: 0.0000						
Distribution: F(31,193)						
Included effects: Entity, Time						

Table 2: Regression Table

Overall we can say that, aging population is the strongest driver of observed provincial mortality differences.

Limitations

We could not access hospital beds per capita, wait times for surgery, and nurses per 100,000 population. These variables would have given a more reliable conclusion about healthcare capacity beyond just physicians and spending. Also provinces with worse health outcomes usually receive more doctors and more government health spending per capita. This makes it difficult to measure the true effect of healthcare supply on mortality. As a result, the coefficients on physicians and spending in our regression are likely biased upward, and the true effect of healthcare may be larger than what we found in my research.

Twenty-three years might not be long enough to see the full impact of healthcare improvements on mortality. Because many healthcare investments, such as screening programs and chronic disease management, take decades to fully reduce mortality. A longer timeline might be able to capture the full improvement in the healthcare sector. Also Canada only has ten provinces, which gives me a small number of cross-sectional units. This limits the statistical power of the fixed effects model and makes it harder to detect small but real effects. and finally the money values in our dataset use different reference years. Median income is in 2023 dollars, while health spending per capita is in 2010 dollars. This does not change the trends within each variable, but it means the two money variables are not on the same price scale in the regression. Because of

this, my results may not be 100% accurate. Using one common base year for all dollar values would give better estimates. although we would get only minor deviation from the actual result.

Conclusion

This paper tested whether provincial differences in treatable mortality across Canada from 2001 to 2023 reflect actual improvements in healthcare performance or changes in population dynamics. Using a two-way fixed effects panel OLS model on 230 province-year observations, I included five independent variables, two healthcare supply proxies (physicians per 100,000 and per capita government health spending) and three socioeconomic and demographic variables (median household income, share of population aged 65 and over, and share of population with postsecondary education). The results show that only the share of population aged 65 and over is marginally significant, while physicians per 100,000, per capita health spending, median income, and postsecondary education are all statistically insignificant after controlling for province and year fixed effects. A one percentage-point increase in the 65+ population share is associated with approximately 10.93 additional treatable deaths per 100,000. This finding is consistent with the increasing Newfoundland–Alberta mortality gap observed in the data, as Newfoundland's 65+ population share is both higher and growing faster than Alberta population.

The insignificance of physicians and spending should not be taken as evidence that healthcare does not matter. Reverse causality is most likely the reason for this result. Because provinces with worse health outcomes tend to receive more doctors and more government health spending per capita, which changes these coefficients upward and hides their true effects. Overall, my evidence suggests that demographic change, specifically population aging, is a stronger driver of provincial mortality differences in Canada than healthcare supply as measured by physician number or real per capita health spending. I can therefore say that simply increasing the number

of doctors or raising health spending may not be enough to close mortality gaps between provinces.

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